

A Nonlinear Low-rank Representation Model with Convolutional Neural Network for Imputing Water Quality Data

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Abstract

Water quality monitoring is a core component of ecological environmental protection. However, due to sensor failure or other inevitable factors, data missing often exists in long-term monitoring, posing great challenges in water quality analysis. This paper proposes a Neural Tucker Convolutional Network (NTCN) model for water quality data imputation, which features the following key components: a) Encode different mode entities into respective embedding vectors, and construct a Tucker interaction tensor by outer product operations to capture the complex mode-wise feature interactions; b) Use 3D convolution to extract fine-grained spatiotemporal features from the interaction tensor. Experiments on three real-world water quality datasets show that the proposed NTCN model outperforms several state-of-the-art imputation models in terms of accuracy.

Index Terms

Water quality data, High-dimensional and sparse, Data imputation, Tensor Neural Network, Embedding representation, 3D Convolution

I. INTRODUCTION

In advancing the modernization drive for harmonious coexistence between humans and nature, water quality monitoring plays an irreplaceable role [1]–[7]. Therefore, accurately collecting water quality-related data is particularly important for making scientific and reasonable work arrangements—it can identify pollution sources, assess ecological risks, and formulate effective response measures [8]–[14]. However, practical data collection faces many challenges [15]–[21]. As the scope of monitoring expands and the number of devices increases, data gaps have emerged. These missing pieces of information prevent us from fully understanding the real state of water quality [22]–[28], so how to accurately fill in these missing data has become an important issue that urgently needs to be addressed [29]–[35].

In recent years, researchers have proposed various methods to address the missing value imputation problem of high-dimensional and sparse data. For example, traditional tensor decomposition methods such as Canonical Polyadic (CP) and Tucker Decomposition (TD) [36]–[42] can only capture the multilinear correlations of data and fail to characterize nonlinear interaction relationships. In neural networks models, NTF can only model the time dimension and is difficult to capture local correlations in the spatial dimension (e.g., the correlation between water quality monitoring stations and water quality indicators) [43]–[49]. Decision trees are employed to model the complex relationships among features and capture nonlinear correlations, thereby enabling accurate data imputation. Nevertheless, this method is prone to the risk of overfitting [50]–[53].

Therefore, we propose the Neural Tucker Convolutional Network (NTCN) model, which combines embedding representation, 3D convolutional operations, and a multi-layer perceptron (MLP) network. It can effectively capture complex nonlinear relationships, thereby improving imputation accuracy. The main contributions of this paper are as follows:

- 1) 3D convolutional feature extraction mechanism. By performing two-layer 3D convolutional operations on the high-order interaction tensor constructed via outer product of embeddings for monitoring stations, water quality indicators, and time slices, it effectively captures the high-order interaction features inherent in the tensor, laying a foundation for accurate imputation.

- 2) Embedding regularization. The embedding weights of monitoring stations, water quality indicators, and time slices are initialized with a uniform distribution, which helps the model converge faster and avoid local optima. Meanwhile, combined with L2 regularization, it reduces the risk of overfitting.
- 3) Comparative experiments based on multiple datasets. Experimental results show that the proposed model outperforms traditional imputation methods in terms of the evaluation metrics RMSE and MSE.

The organizational structure of the remaining sections is as follows: Section II introduces the notation definitions and problem formulation; Section III elaborates on the detailed design of the proposed NTCN model; Section IV describes the experimental setup and the results of comparative experiments; Section V summarizes the full paper and outlines future research directions.

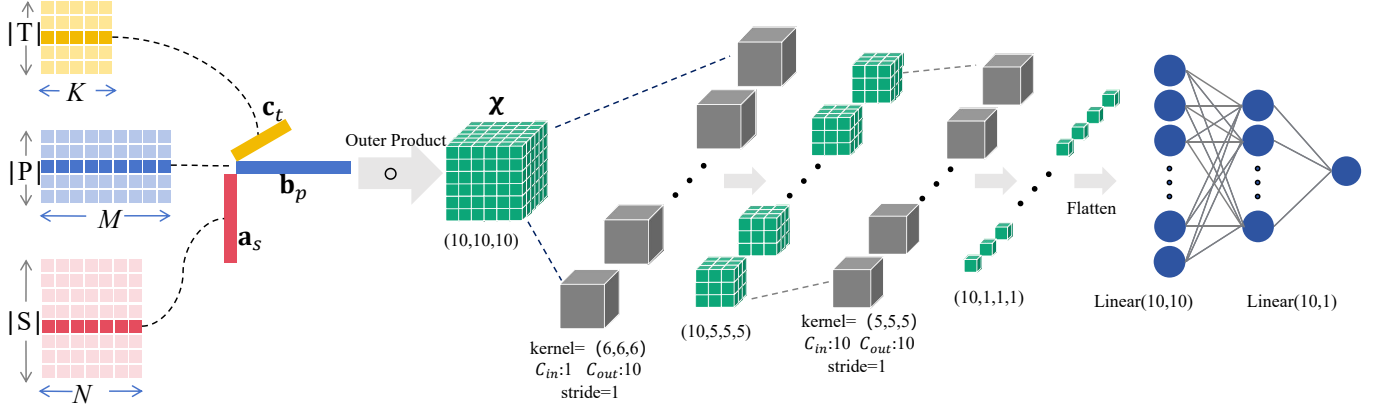


Fig. 1. Architecture of the NTCN model for water quality data imputation. The numerical labels in the figure correspond to the parameter configurations employed in the experiments, including kernel size, stride, and number of channels.

II. PRELIMINARY

A. Notation system

The notations employed in this paper are summarized in Table I.

TABLE I
SYMBOL DESCRIPTION

Symbol	Description
S	Set of monitoring stations, $S = \{s_1, s_2, \dots, s_N\}$
P	Set of water quality indicators (e.g., pH, dissolved oxygen, etc.), $P = \{p_1, p_2, \dots, p_M\}$
T	Set of time slices, $T = \{t_1, t_2, \dots, t_K\}$
$\mathcal{Y} \in \mathbb{R}^{ S \times P \times T }$	Raw water quality tensor, y_{spt} represents the observation on water quality indicator p at station s in time slice t
$\hat{\mathcal{Y}} \in \mathbb{R}^{ S \times P \times T }$	Water quality data tensor after imputation
$\mathbf{a}_s \in \mathbb{R}^N$	Embedding vector of monitoring station s with dimension N
$\mathbf{b}_p \in \mathbb{R}^M$	Embedding vector of water quality indicator p with dimension M
$\mathbf{c}_t \in \mathbb{R}^K$	Embedding vector of time slice t with dimension K
$\mathbf{W}_1 \in \mathbb{R}^{c_1 \times 1 \times k_1 \times k_1 \times k_1}$	Weight of the first 3D convolution layer, c_1 is the number of output channels, k_1 is the kernel size
$\mathbf{W}_2 \in \mathbb{R}^{c_2 \times c_1 \times k_2 \times k_2 \times k_2}$	Weight of the second 3D convolution layer, c_2 is the number of output channels, k_2 is the kernel size
$\mathbf{W}_3 \in \mathbb{R}^{h_1 \times c_{in}}$	Weight of the first layer of the Multilayer Perceptron (MLP), c_{in} is the input dimension, h_1 is the hidden layer dimension
$\mathbf{W}_4 \in \mathbb{R}^{1 \times h_1}$	Weight of the second layer of the MLP
$ \cdot $	The cardinality of a set

B. Problem formulation

The WQD can be represented by an HDS tensor, which is defined as:

Definition 1. (HDS tensor). For a given tensor $\mathcal{Y} \in \mathbb{R}^{|S| \times |P| \times |T|}$, let Ω and Υ denote the observed and unobserved element sets in the tensor, respectively. If $|\Upsilon| \gg |\Omega|$, the tensor $\mathcal{Y}$ is an HDS tensor [54]–[59].

Definition 2. (Tucker Decomposition). TD decomposes into a core tensor multiplied by a factor matrix along each mode, which is formulated as:

$$\mathcal{Y} \approx [\mathcal{G}; \mathbf{A}, \mathbf{B}, \mathbf{C}] = \mathcal{G} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{C}, \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{|S| \times N}$, $\mathbf{B} \in \mathbb{R}^{|P| \times M}$ and $\mathbf{C} \in \mathbb{R}^{|T| \times K}$ are latent factor matrices with ranks of N , M and K , and elements of core tensor $\mathcal{G}$ show the level of interactions among different latent factors. By the definition, the elementwise form of the TD is

$$y_{spt} \approx \hat{y}_{spt} = \sum_{n=1}^N \sum_{m=1}^M \sum_{k=1}^K g_{nmk} a_{sn} b_{pm} c_{tk}. \quad (2)$$

III. THE NTCN MODEL

The overall architecture of the proposed NTCN model is illustrated in Fig. 1, which consists of three core modules: embedding-based high-order interaction tensor construction fusing monitoring stations, water quality indicators and time slices features via outer product, 3D convolutional feature extraction, and MLP prediction.

A. Embedding and Tensor Construction

For the water quality tensor $\mathcal{Y}$, each element is mapped to the interval $[0, 1]$ using the Sigmoid function.

$$y'_{spt} = \sigma(y_{spt}) = \frac{1}{1 + e^{-y_{spt}}}. \quad (3)$$

Through embedding layers, the IDs of monitoring stations, water quality indicators, and time slices corresponding to observed values are mapped into the dense latent feature vectors, thereby addressing the feature representation issue of HDS data:

$$\begin{cases} \mathbf{a}_s = \mathcal{E}(s), \\ \mathbf{b}_p = \mathcal{E}(p), \\ \mathbf{c}_t = \mathcal{E}(t), \end{cases} \quad (4)$$

where $\mathbf{a}_s \in \mathbb{R}^N$, $\mathbf{b}_p \in \mathbb{R}^M$, $\mathbf{c}_t \in \mathbb{R}^K$ and $\mathcal{E}(\cdot)$ denotes the embedding process.

The outer product operation [60]–[65] is used to fuse the embedding vectors of the monitoring stations, water quality indicators, and time slices, constructing a three-order feature tensor to capture the high-order interactions among the three mode [66]–[72]:

$$\mathcal{X}_{spt} = \mathbf{a}_s \circ \mathbf{b}_p \circ \mathbf{c}_t, \quad (5)$$

where $\circ$ denotes the outer product operation, and $\mathcal{X}_{spt} \in \mathbb{R}^{N \times M \times K}$ of which each element is defined as $x_{spt}^{(nmk)} = a_{sn} \cdot b_{pm} \cdot c_{tk}$.

To enhance the robustness on sparse data, the tensor is standardized as:

$$\mathcal{X}'_{spt} = \frac{\mathcal{X}_{spt} - \mu_{\mathcal{X}}}{\sigma_{\mathcal{X}} + \epsilon}, \quad (6)$$

where $\mu_{\mathcal{X}}$ and $\sigma_{\mathcal{X}}$ are the mean and standard deviation of the tensor, respectively, and $\epsilon = 1 \times 10^{-8}$ is a small constant to prevent division by zero.

B. 3D Convolutional Feature Extraction

High-order spatiotemporal interaction features are extracted from the tensor through two layers of 3D convolution:

$$\mathcal{H}_1 = \text{ReLU}(\text{Conv3D}(\mathcal{X}'_{spt}, \mathcal{W}_1)), \quad (7)$$

$$\mathcal{H}_2 = \text{ReLU}(\text{Conv3D}(\mathcal{H}_1, \mathcal{W}_2)), \quad (8)$$

where $\mathcal{H}_1 \in \mathbb{R}^{c_1 \times d_{11} \times d_{12} \times d_{13}}$ and $\mathcal{H}_2 \in \mathbb{R}^{c_2 \times d_{21} \times d_{22} \times d_{23}}$ is the convolutional feature tensor extracted via the respective convolution layer. In this paper, the two 3D convolution kernel have size of $k_1 = 6$ and $k_2 = 5$. The $\text{ReLU}(\cdot)$ denotes the ReLU activation function with the form of $\text{ReLU}(x) = \max(0, x)$.

C. MLP Prediction and Regularization

The convolutional feature tensor $\mathcal{H}_2$ is flattened and then mapped into the predicted value of the water quality indicator via a two-layer MLP structure [73]–[79]:

$$\mathbf{h}_2 = \text{Flatten}(\mathcal{H}_2) \in \mathbb{R}^n, \quad (9)$$

$$\mathbf{h}_3 = \text{ReLU}(\mathbf{W}_3 \mathbf{h}_2 + \mathbf{b}_3) \in \mathbb{R}^m, \quad (10)$$

$$\hat{y}_{spt} = \sigma(\mathbf{W}_4 \mathbf{h}_3 + b_o) \in (0, 1), \quad (11)$$

where $\sigma(\cdot)$ is the sigmoid function which ensures the output lies within the $[0, 1]$, matching with the range of preprocessed water quality data.

The objective function of this model consists of two parts: the mean square error loss (MSE) and L2 regularization. Its specific form is as follow:

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_o + \lambda \cdot \mathcal{L}_{reg}(\Theta) \\ &= \frac{1}{2} \sum_{y_{spt} \in \Omega} (y_{spt} - \hat{y}_{spt})^2 + \lambda \cdot \Theta^2,\end{aligned}\tag{12}$$

where λ is the regularization coefficient, and Θ represents all learnable parameters.

The specific process of the model is presented in Algorithm 1.

Algorithm 1: NTCN Model

Input: $\mathcal{Y}$, c_1, c_2 , k_1, k_2 , λ , T

Output: $\hat{\mathcal{Y}}$

Initialize $\mathbf{a}_s, \mathbf{b}_p, \mathbf{c}_t \sim U(0, 0.004)$, $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3, \mathcal{W}_4$;

$t = 1$;

while $t \leq T$ **do**

for each quadruple $(s, p, t, y_{spt}) \in \Omega$ **do**

$\mathcal{X}_{spt} = \mathbf{a}_s \circ \mathbf{b}_p \circ \mathbf{c}_t$,

$\mathcal{X} \leftarrow$ standardize $\mathcal{X}$ according to (6);

$\mathcal{H}_1 = \text{Conv3D}(\mathcal{X}', \mathcal{W}_1)$,

$\mathcal{H}_2 = \text{Conv3D}(\mathcal{H}_1, \mathcal{W}_2)$,

$\hat{y}_{spt} = \text{MLP}(\mathcal{H}_2)$;

 Backpropagation according to (12);

end

$t = t + 1$;

end

return $\hat{\mathcal{Y}}$

IV. EXPERIMENTAL COMPARISONS

A. Datasets

Based on three sets of real-world water quality data from 24 monitoring stations in Victoria Harbour established by the Environmental Protection Department of Hong Kong: surface layer D1 with 129,406 records, middle layer D2 with 121,218 records, and bottom layer D3 with 129,415 records. This study covers 24 water quality-related parameters over 903 time slices. The datasets are divided into training set (Γ), validation set (Λ), and test set (Φ) sets at a ratio of 1:2:7 to evaluate the performance of the proposed model.

B. Evaluation metrics

To measure the model's performance, this study employs the common Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) as evaluation metrics [80]–[87]:

$$\text{MAE} = \frac{1}{|\Phi|} \sum_{y_{spt} \in \Phi} |y_{spt} - \hat{y}_{spt}|,\tag{13}$$

$$\text{RMSE} = \sqrt{\frac{1}{|\Phi|} \sum_{y_{spt} \in \Phi} (y_{spt} - \hat{y}_{spt})^2}.\tag{14}$$

C. Comparison Results

We compare the results of the proposed NTCN model with four state-of-the-art imputation models on three datasets in terms of imputation accuracy [88]–[94]. The tested models are: a) M1: The proposed NTCN model; b) M2: A biased low-rank tensor model based on non-negative multiplicative updates; c) M3: An imputation model that employs the Cauchy loss function and handles outliers without explicit outlier detection; d) M4: A multi-dimensional prediction model based on tensor decomposition and the reconfiguration optimization algorithm; e) M5: A model adopting a tensor completion mechanism that combines CP decomposition, standard gradient descent and alternating least squares optimization.

To evaluate its performance, the proposed model is benchmarked against four current best-performing models across three datasets, with a focus on imputation accuracy.

TABLE II
IMPUTATION PERFORMANCE COMPARISON

Dataset		M1	M2	M3	M4	M5
D1	RMSE	0.0243	0.0256	0.0260	0.0273	0.0267
	MAE	0.0149	0.0162	0.0148	0.0162	0.0163
D2	RMSE	0.0259	0.0262	0.0266	0.0287	0.0270
	MAE	0.0168	0.0175	0.0161	0.0177	0.0176
D3	RMSE	0.0249	0.0258	0.0262	0.0277	0.0259
	MAE	0.0164	0.0176	0.0159	0.0175	0.0166

The rank of all models are set to 10 and the iteration rounds are set to 1000. Note that the tested models are terminated if the error in two consecutive iteration rounds is less than 10^{-5} . Tables II present the statistical results, and it can be observed that:

- 1) The NTCN model significantly outperforms other tested models in terms of imputation accuracy. As shown in Table II, on the D1 dataset, the NTCN model M1 achieves an RMSE of 0.0243 and an MSE of 0.0149. Compared with M2, which has an RMSE of 0.0256 and MSE of 0.0162; M4, which has an RMSE of 0.0273 and an MSE of 0.0162; and M5, which has an RMSE of 0.0267 and MSE of 0.0163, the MSE of M1 is reduced by 8.02%, 8.02%, and 8.59% respectively. It only differs from M3's MSE of 0.0148 by 0.0001, being basically the same and maintaining the optimal level.
- 2) On the D2 dataset, the RMSE of M1 is significantly lower than that of other models, and the MAE of M1 is 0.0168. Compared with M2's MAE of 0.0175, M4's MAE of 0.0177, and M5's MAE of 0.0176, it is reduced by 4.00%, 5.08%, and 4.55% respectively. It is only slightly higher than M3's 0.0161, and still in the leading echelon overall.
- 3) On the D3 dataset, the MAE of M1 is 0.0164. Compared with M2's MAE of 0.0176, M4's MAE of 0.0175, and M5's MAE of 0.0166, it is reduced by 6.82%, 6.29%, and 1.20% respectively. It only differs from M3's 0.0159 by 0.0005, maintaining advantages as well. In terms of RMSE, M1 is the lowest among the three datasets, fully reflecting the absolute advantage of the model in imputation accuracy.

In summary, the proposed NTCN model achieves high accuracy in imputing missing values of water quality data. It can accurately capture the high-order interaction relationships among monitoring stations, water quality indicators, and time slices, making both RMSE and MAE better than those of most comparison models, thus providing an efficient and reliable solution for practical water quality monitoring data processing.

V. CONCLUSIONS

To achieve accurate imputation of missing data, this paper proposes a 3D convolution-based model. Specifically, the model maps the features of monitoring stations, water quality indicators, and time slices to a low-dimensional latent space through embedding layers, and constructs a 3D feature tensor using outer product operations to capture the high-order interaction relationships among the three. On this basis, the model adopts two-layer 3D convolutional operations for feature extraction and dimensionality reduction of the tensor. Leveraging the local perception capability of convolutional kernels, it mines local correlation patterns in the latent space, and then completes nonlinear mapping and final prediction through a MLP. The advantages of the model are as follows: on the one hand, it reduces the scale of model parameters while retaining key interaction information, effectively alleviating the overfitting risk caused by HDS data; on the other hand, the 3D convolutional operation can capture the complex correlations of features among the monitoring station, water quality indicator, and time slice. Performance evaluation on water quality datasets shows that the proposed model outperforms existing imputation models in missing data imputation tasks [95]–[100]. In the future, we plan to explore on air quality monitoring datasets and verify its application potential in broader ecological sensing tasks.

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